Marketing Analytics

Conjoint and Product Design

By Nathan Yang

Session 05



A Subscription-Bundle Decision

- A firm can change quality, delivery, and monthly price.
- Customers must trade off feasible attribute combinations.
- The choice set includes the option to buy nothing.

Choice-Task Design

- Vary price independently of features where feasible.
- Use realistic combinations and clear descriptions.
- Retain respondent identity across repeated tasks.

Utility and Price

Systematic utility combines product features and price.

$$V_{ij} = \beta_q q_j + \beta_f f_j + \beta_p p_j$$

Utility has an arbitrary scale. Price is measured in dollars per billing period.

Logit Choice Probabilities

Normalize probabilities over the alternatives available to customer i.

$$P_{ij} = \frac{\exp(V_{ij})}{\sum_{k \in \mathcal{J}_i} \exp(V_{ik})}$$

The denominator includes the outside option and every available product.

Willingness to Pay

Under linear price utility, equate a feature gain with a price increase.

$$\mathrm{WTP}_f = -\frac{\beta_f}{\beta_p}$$

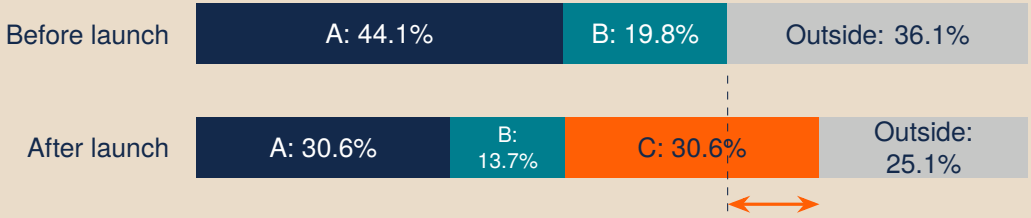
If $\beta_f = 1.2$ and $\beta_p = -0.1$ per dollar, the tradeoff is 12 dollars.

Two Bundles and an Outside Option

Option	Quality	Price	Utility
A	Premium	10	0.2
B	Standard	6	-0.6
Outside	None	0	0

Synthetic inputs: quality coefficient 1.2 and price coefficient -0.1.

A Launch Redistributes Existing Sales



Synthetic logit model. New bundle C has A's modeled attributes.

Incremental Contribution

- Attach contribution margins to each simulated purchase.
- Compare the full product portfolio before and after launch.
- Deduct development and operating costs separately.

A Probability Check

Calculate the synthetic baseline shares.

```
from math import exp
utilities = {'A': 0.2, 'B': -0.6, 'Outside': 0.0}
weights = {k: exp(v) for k,v in utilities.items()}
shares = {k: v/sum(weights.values())
          for k,v in weights.items()}
assert abs(sum(shares.values()) - 1) < 1e-12
```

A has about 44.1% share, B 19.8%, and the outside option 36.1%.

Substitution Assumptions

- A close copy can draw mostly from its nearest competitor.
- Basic logit instead preserves relative odds among existing alternatives.
- Holdout choice tasks can reveal systematic substitution errors.

Coding Exercise: A Product Launch

- Add C to the synthetic profile table.
- Calculate portfolio share and contribution before and after launch.
- Compare C’s share with the change in category participation.
- Check a case with no outside option.

Validation and Market Transfer

- Hold out respondents or tasks according to the intended use.
- Check coding, price units, and probabilities.
- Relate stated preference to awareness, distribution, and availability.

mHealth Product Economics

- Food logs reveal recorded use and category histories.
- A paid-plan choice model needs additional price and preference evidence.
- Financial scenarios can state the missing inputs explicitly.

AI-Assisted Conjoint Analysis

- Provide the choice-set identifier and the selected alternative.
- Require probabilities to sum to one within each task.
- Preserve respondent groups in validation.
- Request a portfolio comparison with explicit margins.

Price Presentation and Customer Understanding

- A base price and a mandatory fee determine the total paid.
- A stated-choice task can reveal different responses to equivalent totals.
- The feasible pricing design depends on the applicable rule and industry.

Product-Design Discussion

- How much of C’s demand comes from existing products?
- Which design change would identify the price tradeoff?
- What evidence could change the launch recommendation?